

The Energy Cost of Prompting a Large Language Model

Abigail Pitcairn - University of Southern Maine - abigail.pitcairn@maine.edu

Paper &
Github:



Motivation & Goal

Problem: AI energy research focuses on training and inference costs, but what about the cost of everyday use?

Research Questions:

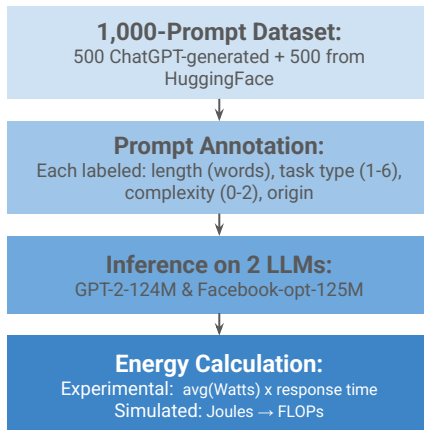
RQ1 – Average energy cost of one LLM prompt?

RQ2 – Which prompt qualities drive energy use?

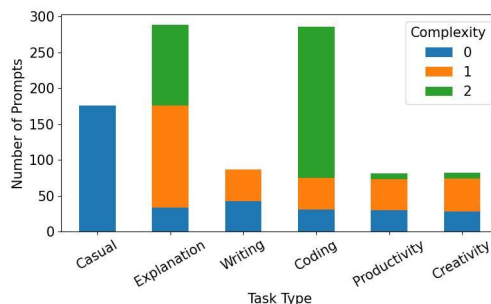
Energy Comparison:

Activity	Energy
1 GPT-2 prompt	0.09 Wh
1 Google search	0.30 Wh
LED bulb, 1 min	1.00 Wh
LED TV, 30 min	30.00 Wh

Dataset & Methodology



Distribution of Task Type and Complexity



Key Results

0.087 Wh

avg energy /
GPT-2 prompt

0.114 Wh

median energy /
GPT-2 prompt

Energy Cost Range:

0.005 - 0.154 Wh

Simulated results align closely with experimental results.

Conclusion & Future Work

Input & Output Length: significantly more energy cost for prompts and outputs with more words.

Limitations: only tested models with $\geq 125M$ parameters.

Future Work: analyze output quality and larger models.

Average Total Energy Consumption per Prompt, Sorted Lowest to Highest

